Gian Carlo D Trilles BSCpE 3A

TE

Assignment

7. $f(w,x,y,z) = \Pi(6,8,9,10,11,12,13)$

wx \ yz	00	01	11	10
00	1	1	1	1
01	1	1	1	0
11	0	0	1	1
10	0	0	0	0

$SDP = w'y' + w'z' + w'x' + wxy$

$POS = (w'+y)(w'+x)(w+x'+y'+z)$

DATE

2. $f(w,x,y,z) = \Sigma(2,3,5,7,10,11,13,14,15$

wx \ yz	00	01	11	10
00	0	0	1	1
01	0	1	1	0
11	0	1	1	1
10	0	0	1	1

SOP = $xz + wy + x'y$

POS = $(x+y)(w+x'+z)(y+z)$

DATE

3. $f(w,x,y,z) = \Sigma(0,4,5,10,11,13,15)$

show all possible answers

wx \ yz	00	01	11	10
00	1	0	0	0
01	1	1	0	0
11	0	1	1	0
10	0	0	1	1

$SOP = wxz + wx'y + w'xy' + w'y'z'$

$POS = (w'+x'+z)(w+y')(w'+x+y)(w+x+z')$

wx \ yz	00	01	11	10
00	1	0	0	0
01	1	1	0	0
11	0	1	1	0
10	0	0	1	1

$SOP = xy'z + wyz + w'y'z' + wx'y$

$POS = (x'+y'+z)(x+y+z')(w'+y+z)(w+y')$

Signature

DATE

4. $f(a,b,c,d) = \Sigma(4,6,13,15) + d(1,7,8,14)$

wx \ yz	00	01	11	10
00	0	–	0	0
01	1	0	–	1
11	0	1	1	–
10	–	0	0	0

SOP = $abd + a'bd'$

POS = $(a + d')(b)(a' + d)$

DATE

5. $f(a,b,c,d,e) = \Sigma(1,2,8,9,15,17,24,25,27) + d(0,21,31)$

ab \ cde	000	001	011	010	110	111	101	100
00	–	1	0	1	0	0	0	0
01	1	1	0	0	0	1	0	0
11	1	1	1	0	0	–	0	0
10	0	1	0	0	0	0	–	0

$SOP = a'b'c'e' + abc'e + bc'd' + c'd'e + bcde$

$POS = (a'b+c)(b'+d'+e)(c'+d)(c'+e)(b+d'+e')(a+c+d'+e')$